CS1851 HW3 Final Reflection

Data and preprocessing:
The heart disease dataset has 1025 samples and 13 features. It had no missing values but after removing duplicate values, the dataset was reduced to only 302 samples. The class balance is relatively good with around 140 negative and 160 positive samples. Non categorical features for the heart disease dataset appear roughly normally distributed without major outliers. Some categorical variables like sex are more skewed than others (such as chest pain and exercise-induced angina). Assuming common sex label convention for kaggle datasets, there are two times more females than males in the heart disease dataset. The biological aging dataset has 37 samples and 482,421 features after extracting only methylation features. Unlike the heart disease dataset, some columns had missing values, which were filled using mean imputation (though a column with only NaN values was dropped). The severe imbalance between samples and features suggests that overfitting may occur on this dataset. Given the size of the feature dataset, I performed topK feature selection based on variance. Notably, there was no apparent difference in results between topK=80 and topK=1000 on performance metrics. Standard scaling was applied for both models.

Modeling and hyperparameters / Metrics and Visuals:
For both the classification and regression tasks using the heart disease dataset and biological aging dataset, respectively, I tested different model kernels and values for parameters C, degree, and gamma. For the classification task, the best overall performing (tuned) model across the performance metrics accuracy, precision, recall, f1, and roc_auc was a model with the parameters kernel=poly, C=1, degree=3, and gamma=0.1. Compared to the base model (kernel=linear, C=1, and gamma=scale), this tuned model performed better across all metrics, though roc_auc was comparable (Table 1). An alternative model with slightly worse overall metrics but better recall is one with parameters kernel=rbf, C=1, gamma=0.1. The difference in performance metrics are observable by inspection when comparing the classification matrices of the base and tuned model–with the tuned model correctly classifying more cases. (Figure 1).

		Accuracy	Precision	Recall	F1	ROC_AUC
Classification (Heart Disease Dataset)	Base (linear, 1, scale)	0.770492	0.727273	0.827586	0.774194	0.862069
	Tuned (poly, 1, 3, 0.1)	0.836066	0.787879	0.896552	0.83871	0.871767
	Alternative (rbf, 1,2, 0.1)	0.819672	0.736842	0.965517	0.835821	0.883621
		MAE	RMSE	R^2		
Regression (Biological Aging dataset)	Base (linear, 1, scale)	9.264621	10.839988	-2.617288		
	Tuned (rbf, 10, 0.001)	4.470919	5.558529	0.048858		
	Alternative (poly, 0.1, 4, 0.001)	3.925001	5.723635	-0.008485		

Table 1. Performance metrics of classification and regression models.

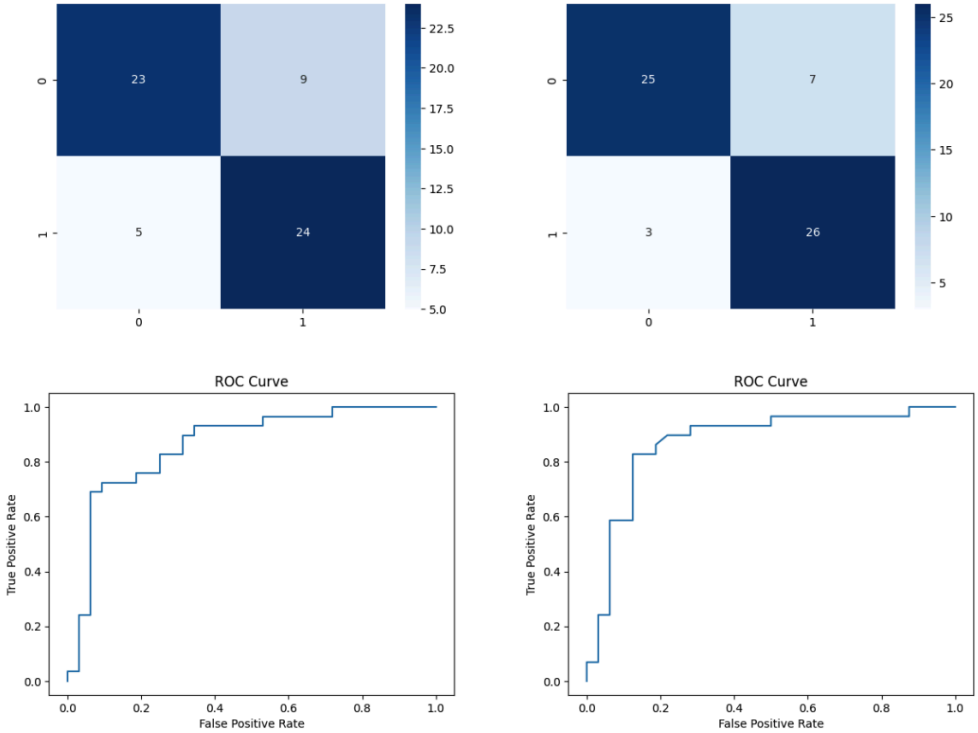

Figure 1. Confusion matrices and ROC curves for base and tuned model. Base model figures on left and tuned model figures on right.

Polynomial and RBF kernels allow nonlinear classification (visually this means in a 2D plot its boundary can be curved and nonlinear). This allows the models to classify more complex data than a linear kernel which are limited to linearly separable data and have straight boundary lines, so it's reasonable that both the polynomial and RBF kernel based models performed the best on this task (Figure 2). In general, RBF kernels are better at working with nonlinear data because it maps the data into an infinite-dimensional space.

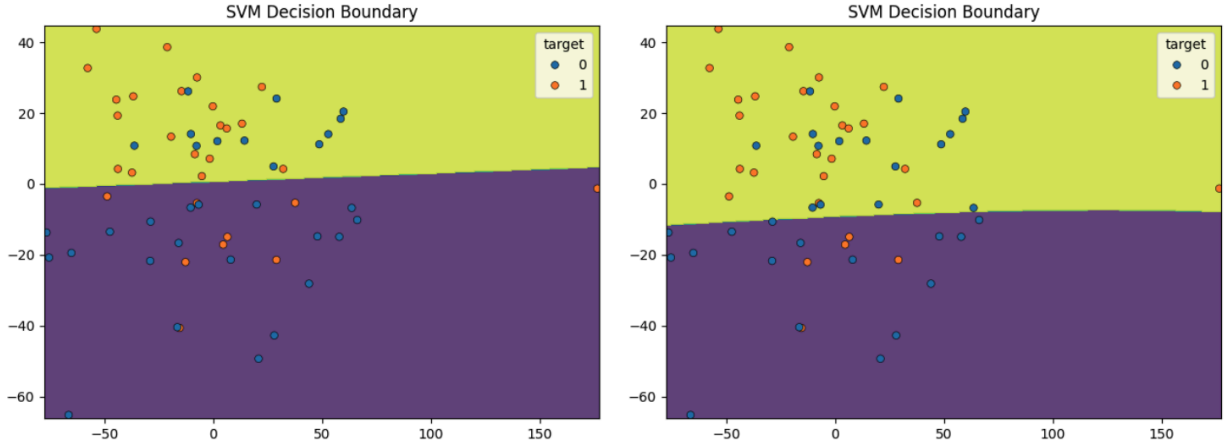

Figure 2. Decision boundary plots for linear (left) and rbf (right) kernels.

The parameter C represents the weighting between the svm margin (distance between boundary line and support vectors) and slack (data points within margin or on the wrong side of boundary). For example, a low C indicates a wider margin and greater tolerance for data points on the wrong side of the boundary line. In the decision boundary plots, the curve will also appear more smooth. Thus, during hyperparameter tuning models with a lower C may perform better, although values that are too low can lead to underfitting and may not be desirable depending on the type of data in the dataset so C=1 is reasonable for the best performing models (Figure 3).

For a poly kernel, degree controls the flexibility of the decision boundary. Higher degrees allow the model to capture more complex boundaries but may lead to overfitting. A moderate value such as degree=3 observed in the best model, allows a balance between capturing nonlinear structure and not overfitting the data (Figure 3).

Gamma influences how much individual data points influence the shape of the decision boundary, with larger gamma values making the model more sensitive to each data point, resulting in more complex decision boundaries. In the poly kernel, increasing gamma likely led to increased model complexity resulting in better performance (Figure 3). For the rbf kernel, given that it already has lots of flexibility in its boundaries, it makes sense that a gamma that balances between providing complexity without overfitting may perform the best (Figure 3).

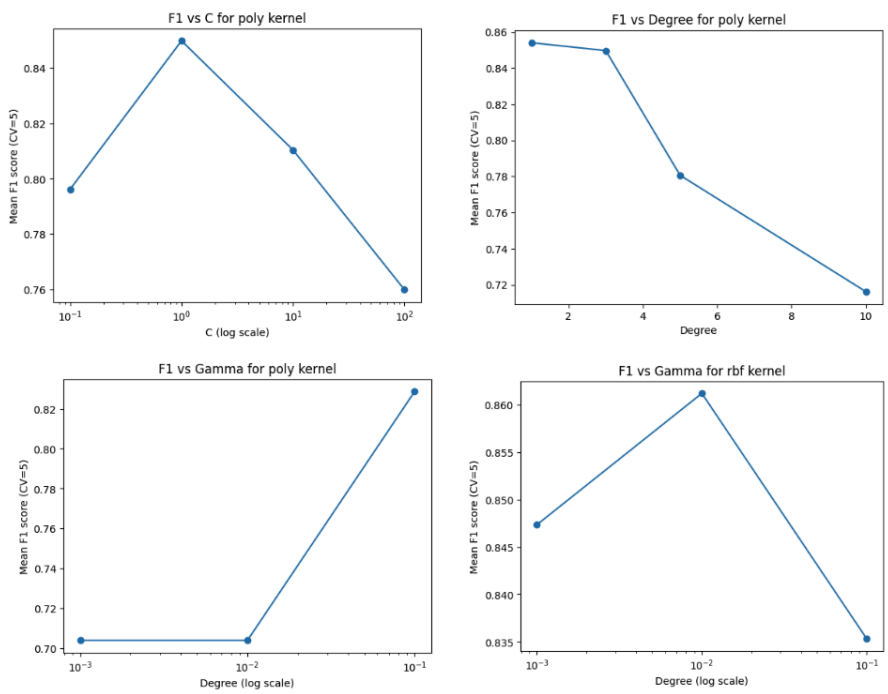

Figure 3. Hyperparameter tuning on poly and rbf kernel with F1 metric

For the regression task, the model with kernel=rbf, C=10, and gamma=0.001 performed the best. Its MAE and RMSE metrics appear much lower than the base model and I suspect it's because of the kernel type (Figure 1). Given the lack of samples (only 37), the high degree but low gamma suggests the dataset is easier to classify as the model has little tolerance for

misclassified or data points that violate the margin and reduced model complexity. Though a model with kernel=poly, C=0.1, degree=4, and gamma=0.001 performed comparably across MAE and RMSE metrics, it had an negative r^2 value (indicating worse than simply predicting mean) while the rbf kernel model had a r^2 of 0.05 (Figure 1). This r^2 value is low and suggests that the data has a lot of variance and/or the model is not very predictive. The parity plot of the tuned model performing the regression task suggests that the performance is okay, with one main outlier (Figure 4). The corresponding residuals plot reflects this observation with most residuals being within around ±4 and an outlier (residual > 15.0) (Figure 4).

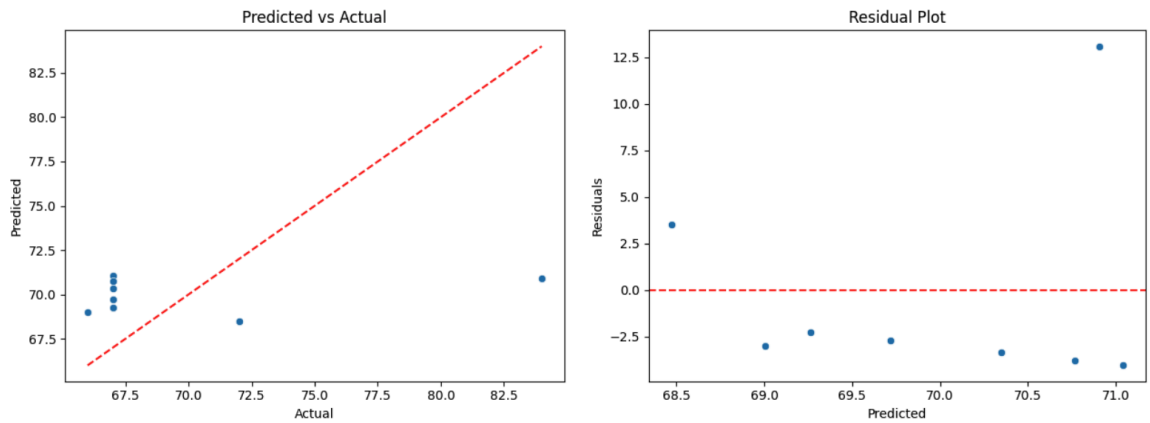

Figure 4. Parity plot (left) and residual plot (right) of regression task on biological aging dataset

Sensitivity:

Scaling was used for all models and in general, it helps improve the models predictive power by scaling the features to mean 0 and variance of 1. This helps prevent some features from dominating others through magnitude alone. Compared to the base classification model, a model with the same hyperparameters with no scaling applied to its data performed worse across all performance metrics by around 1-2% points for each metric. Applying PCA improved the base classification model across all metrics as well, increasing each metric by around 4% points, though the metrics were still not better than the tuned model and PCA had little effect on the performance metrics of the tuned model. When applying PCA to the classification model, n_components=0.95 was passed into the argument for PCA, so that the retained components capture 95% of the variance in the data. When applying PCA to the regression model, n_components=0.95 had little impact on the performance metrics so I tested n_components=5, which retains 5 principal components. This improved performance of the base model by what appears to be significantly, halving the MAE and RMSE. Improvement in the MAE metric was observed when applied to the tuned model but it was not as dramatic and RMSE did not improve. I also wanted to note that the models that used gamma="scale" generally performed better when PCA was added to the pipeline than the tuned gamma values on the heart disease dataset.

Changing degree and gamma for polynomial kernels exhibited a clear trend for the heart disease and biological aging dataset–performance (F1 and negative RMSE) decreased as the parameter was increased (Figure 5). This indicates that a more basic poly-kernel model can perform well on the aging dataset. It also suggests that linear kernels (which is essentially a polynomial kernel with degree = 1) could work just fine on both datasets (Figure 3 and Figure 5).

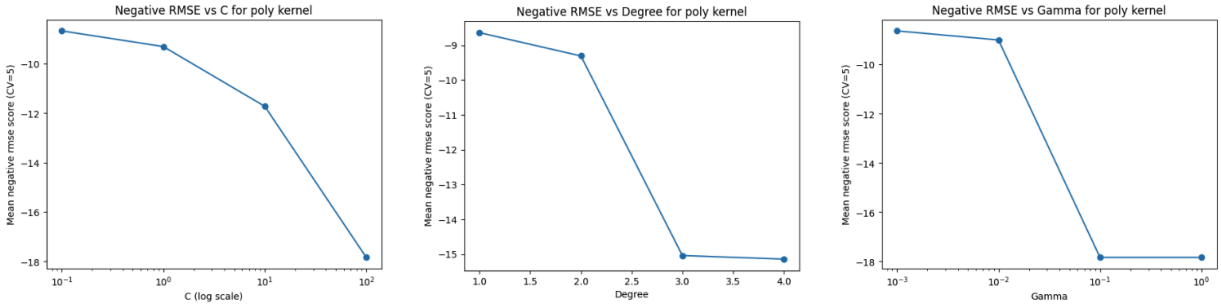

Figure 5. Performance metrics vs degree for poly kernels.

I also compared “scale” and “auto” in previous analyses, finding little to no quantitative difference between the two.